INCH-POUND

MIL-PRF-6106/9E
22 September 2011
SUPERSEDING
MIL-PRF-6106/9D
10 November 2000

PERFORMANCE SPECIFICATION SHEET

RELAY, ELECTROMAGNETIC, CONTINUOUS DUTY, 60 AMP, 3PDT, ENVIRONMENTALLY SEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

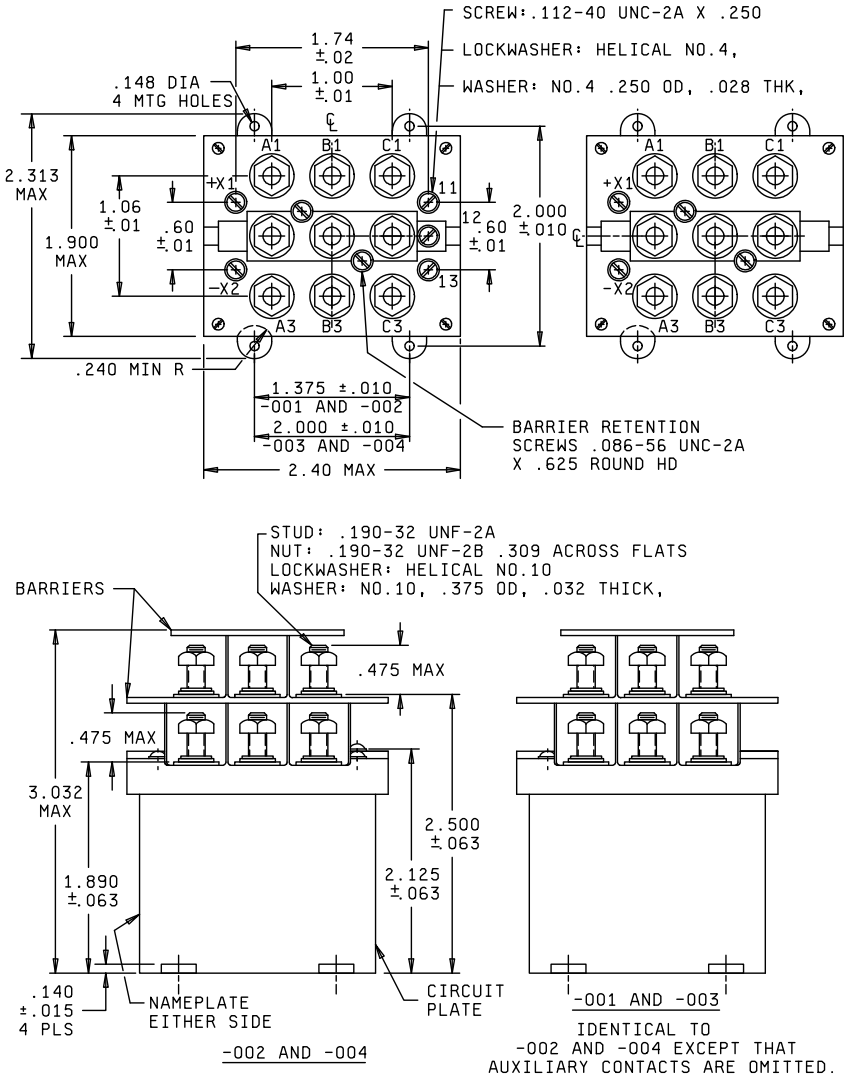
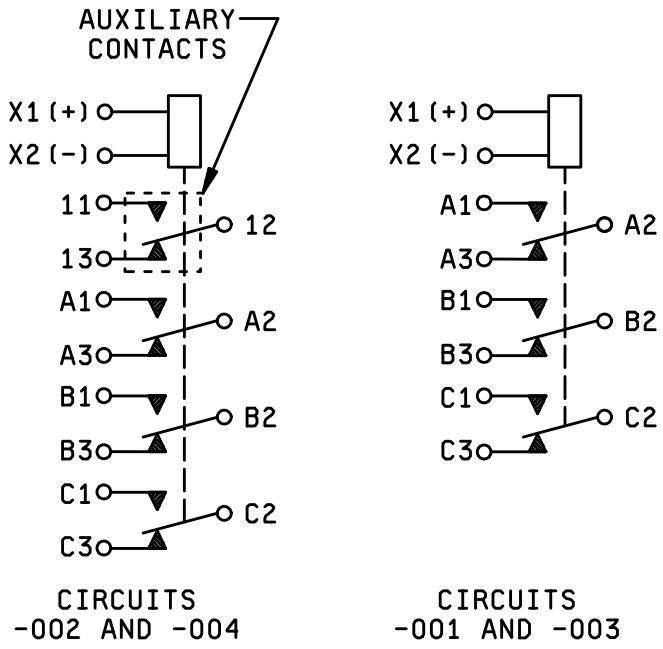


FIGURE 1. Configurations and dimensions.

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Inches	mm	Inches	mm	Inches	mm
.010	0.25	.309	7.86	1.890	48.01
.02	0.51	.375	9.53	1.900	48.26
.028	0.71	.440	11.18	2.000	50.80
.032	0.81	.600	15.24	2.115	53.72
.063	1.60	.625	15.88	2.313	58.75
.148	3.76	1.000	25.40	2.40	60.96
.186	4.72	1.060	26.92	2.500	63.50
.240	6.10	1.375	35.18	3.032	77.01
.250	6.35	1.74	44.20		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.031 (0.79 mm).
4. Weights include cover, barriers, and terminal hardware.
5. When using relay at rated 60 amps, the use of lugs per Burndy part number YAEV6C-G8 or equal is recommended.
6. Terminal temperature rise under continuous current conditions is 95°C maximum.

FIGURE 1. Relay, outline drawing - Continued.

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TABLE I. Dash number and operating characteristics.

Dash no.	Type	Coil data												Time milliseconds-maximum					
		Coil	Nominal			Maximum			Max pickup voltage 1/			Hold voltage 1/	Drop out voltage 1/	Operate 2/	Release 3/	Bounce			
			Volts	Freq	Res	Volts	Ampere		Normal 1/	High temp test	Cont current test					Main		Auxiliary	
							Inrush	Steady state								NO	NC	NO	NC
-001	1	X1, X2	28	DC	-	29	1.25	.25	17	20	21.5	7.0	1.5	25	25	2	2	-	-
-002	1	X1, X2	28	DC	-	29	1.25	.25	17	20	21.5	7.0	1.5	25	25	2	2	2	2
-003	1	X1, X2	115	400	-	124	.71	.1	85	95	100	40	10	25	55	2	2	-	-
-004	1	X1, X2	115	400	-	124	.71	.1	85	95	100	40	10	25	55	2	2	2	2

1/ Over the temperature range 2/ With nominal coil voltage 3/ From nominal coil voltage

TABLE II. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles $x10^3$	28 V dc 1/				115 V ac, 1 phase 1/				115/200 V ac, 3 phase 1/			
		Main		Auxiliary		Main		Auxiliary		Main		Auxiliary	
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz
Resistive	100	-	-	3	3	60	-	3	-	60	-	-	-
Inductive	100	-	-	1.5	1.5	60	-	1.5	-	60	-	-	-
Motor	50	-	-	-	-	40	-	-	-	40	-	-	-
Lamp		-	-	.5	.5		-	.5	-		-	-	-
Transfer load 2/		-	-				-		-		-	-	-
Mechanical life reduced current	200	-	-	.75	.75	15	-	.75	-	15	-	-	-
Mixed loads 3/	50	-	-	Per spec		4	-	-	-	4	-	-	-

1/ Absence of value indicates relay is not rated for that application.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating.
3/ Mixed loads to be conducted at +71°C. Number of mixed loads cycles following continuous current test - 5000.

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REQUIREMENTS

Weight: 001 and 002 - 10.5 ounces maximum.
003 and 004 - 11.5 ounces maximum.

Temperature range: -55°C to +71°C.

Maximum altitude rating: 50,000 feet 1/.

Shock g-level: ½ sine, 25 g's.

Duration: 6 to 9 ms.

Maximum duration contact opening: 1 ms.

Vibration-sinusoidal: See table III.

TABLE III. Vibration-sinusoidal.

Dash No.	Vibration level					
	5-10 Hz	10-55 Hz	55-400 Hz	400-800 Hz	800 - 2000 Hz	
					-55°C and 25°C	+71°C
-001	.08 DA	.06 DA	10 g's	10 g's	10 g's	7 g's
-002			10 g's	8 g's	8 g's	7 g's
-003			10 g's	10 g's	10 g's	7 g's
-004			10 g's	8 g's	8 g's	7 g's

Vibration (random): Not applicable.

Acceleration: 15 g's/

Seal:

Applicable specification: MIL-STD-202, method 112, test condition C, procedure IV.

Leak rate: $6x10^{-4}$ STD cc/s.

Strength of terminals and mounting studs:

Torque:

Main terminals: 24 inch-pounds maximum, 18 inch-pounds recommended installation torque.

Auxilary terminals: 4.4 inch-pounds maximum, 3.3 inch-pounds recommended installation torque.

Insulation resistance, initial: 200 megohms minimum.

After life or environmental tests: 100 megohms minimum.

1/ Application range can be extended to 80,000 feet by suitably encapsulating the terminals.Dielectric withstanding voltage (sea level) 2 to 5 seconds.

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	Initial		After life tests	
	28 V dc	115 V ac	28 V dc	115 V ac
Coil to case	1250	1500	1000	1125
Auxiliary contacts	1250	1250	1000	1000
Across open power contacts	1250	1250	625	625
All other points	1800	1800	1350	1350

Dielectric withstanding voltage (altitude) - 1 minute.

	28 V dc	115 V ac
Coil to case	500	500
Auxiliary contacts	500	500
All other points	700	700

Maximum contact drop, initial .150 volt.

After life test .200 volt.

	NO (V ac)	NC (V ac)
Overload current	320	320
Rupture current	400	400

Duty rating: Continuous.

RFI specification (applicable to coil circuits of ac operated relays): MIL-STD-461.

Part number M6106/9- (dash number from table I).

Qualification by similarity: See MIL-PRF-6106.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

1/ Number of rupture cycles after mixed loads test - 5.

Referenced documents. In addition to MIL-PRF-6106, this document references the following:

MIL-STD-202 MIL-STD-461

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Custodians:
Air Force - 85
DLA - CC

Review activities:
Air Force – 99

Preparing activity:
DLA - CC

(Project 5945-2011-059)

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